How to find the standard matrix of a linear transformation without given the formula?

Answer:

When you don't have an explicit formula like T(x, y) = (2x, x-y), you must rely on the fundamental definition of the standard matrix and the properties of linear transformations.

The core principle is this:

The columns of the standard matrix for a linear transformation T are the images of the standard basis vectors under T.

Let's break this down into a practical method.

A linear transformation T: $\mathbf{R}^n \rightarrow \mathbf{R}^m$ is completely determined by what it does to the basis vectors of its domain, $\mathbf{R}^n$. The standard basis vectors in $\mathbf{R}^n$ are $\mathbf{e_1}, \mathbf{e_2}, \ldots, \mathbf{e_n}$, where $\mathbf{e}_i$ is the vector with a 1 in the i-th position and 0s everywhere else.

For example, in $\mathbf{R}^2$, the standard basis vectors are: $\mathbf{e_1} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \mathbf{e_2} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Any vector **x** in $\mathbf{R}^2$ can be written as a linear combination of these basis vectors:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = x_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + x_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = x_1\mathbf{e_1} + x_2\mathbf{e_2}$$

Because T is a linear transformation, we can apply the linearity condition:

$$T(\mathbf{x}) = T(x_1\mathbf{e_1} + x_2\mathbf{e_2}) = x_1T(\mathbf{e_1}) + x_2T(\mathrm{e}_2)$$

This last expression is just a matrix-vector product as follow:

If we define a matrix A whose columns are the vectors T($\mathbf{e_1}$) and T($\mathbf{e_2}$):

$$\mathbf{A} = \begin{bmatrix} | & | \\ T(\mathbf{e_1}) & T(\mathbf{e_2}) \\ | & | \end{bmatrix}$$

Then the product **Ax** is: $\mathbf{Ax} = \begin{bmatrix} | & | \\ T(\mathbf{e_1}) & T(\mathbf{e_2}) \\ | & | \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = x_1T(\mathbf{e_1}) + x_2T(\mathbf{e_2})$

Since T(**x**) = **Ax**, the matrix **A** is the standard matrix for T.

Now to find the standard matrix if you aren't given the formula, you need enough information to figure out where a complete set of basis vectors of the domain $\mathbf{R}^n$ land under the transformation. This information usually comes in one of two forms:

1. **Images of the standard basis vectors:** You are told what the transformation does to the set of standard basis vectors $\mathbf{e_1}, \mathbf{e_2}, \ldots, \mathbf{e_n}$. We can form a matrix **A** by placing these images T($\mathbf{e_1}$), T($\mathbf{e_2}$), …, T($\mathbf{e_n}$) as columns in that order. Then **A** will be the standard matrix of the linear transformation T.
2. **Images of a non-standard basis vectors:** You are told what the transformation does to a set of non-standard basis vectors $\mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_n}$. There are two ways to find the standard matrix of the linear transformation T:
   i. Use the images T($\mathbf{v_1}$), T($\mathbf{v_2}$), …, T($\mathbf{v_n}$) to derive the images T($\mathbf{e_1}$), T($\mathbf{e_2}$), …, T($\mathbf{e_n}$) of the standard basis vectors and follow with part 1 above.

ii. "Stacking method"- Form a matrix **B** by placing the basis vectors $\mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_n}$ as columns; and form a matrix **C** by placing the respective images T($\mathbf{v_1}$), T($\mathbf{v_2}$), ..., T($\mathbf{v_n}$) as columns:

$$\mathbf{B} = \begin{bmatrix} | & | & & | \\ \mathbf{v_1} & \mathbf{v_2} & \ldots & \mathbf{v_n} \\ | & | & & | \end{bmatrix} \text{ and } \mathbf{C} = \begin{bmatrix} | & | & & | \\ T(\mathbf{v_1}) & T(\mathbf{v_2}) & \ldots & T(\mathbf{v_n}) \\ | & | & & | \end{bmatrix}$$

$$\text{Then } \mathbf{C} = \begin{bmatrix} | & | & & | \\ \mathbf{Av_1} & \mathbf{Av_2} & \ldots & \mathbf{Av_n} \\ | & | & & | \end{bmatrix} = \mathbf{A}\begin{bmatrix} | & | & & | \\ \mathbf{v_1} & \mathbf{v_2} & \ldots & \mathbf{v_n} \\ | & | & & | \end{bmatrix} = \mathbf{AB}.$$

Since we know **B** and **C**, and **B** is non-singular (its columns are linearly independent). we can find $\mathbf{A} = \mathbf{CB}^{-1}$.

Scenario 1: **Standard basis vectors**

Let's say T: $\mathbf{R}^2 \rightarrow \mathbf{R}^2$ is a linear transformation that maps (1, 0) to (0, -1) and maps (0,1) to (1,1).

Since $\mathbf{e_1}$ = (1, 0) and $\mathbf{e_2}$ = (0, 1) are the standard basis vectors for $\mathbf{R}^2$, the given information means T($\mathbf{e_1}$) = (0, -1) and T($\mathbf{e_2}$) = (1, 1). Place these images as columns in our matrix **A**:

$$\mathbf{A} = [T(\mathbf{e_1}) \quad T(\mathbf{e_2})] = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}$$

This is the standard matrix of our linear transformation T.

Scenario 2: **Non-standard basis vectors**

Let's say T: $\mathbf{R}^2 \rightarrow \mathbf{R}^2$ is a linear transformation such that $T\begin{pmatrix}1\\2\end{pmatrix} = \begin{pmatrix}3\\3\end{pmatrix}$ and $T\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}1\\2\end{pmatrix}$.

Method 1: Find T($\mathbf{e_1}$) and T($\mathbf{e_2}$)

1. **Goal:** Find T($\mathbf{e_1}$) and T($\mathbf{e_2}$). We already have T($\mathbf{e_2}$) = (1, 2). We need to find T($\mathbf{e_1}$).
2. **Express $\mathbf{e_1}$ as a linear combination of the vectors whose images we know.**
   Let $\mathbf{v_1}$ = (1, 2) and $\mathbf{v_2}$ = (0, 1). We want to find scalars c_1 and c_2 such that:
   $\mathbf{e_1} = c_1\mathbf{v_1} + c_2\mathbf{v_2}$
   $\begin{pmatrix}1\\0\end{pmatrix} = c_1\begin{pmatrix}1\\2\end{pmatrix} + c_2\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}c_1\\2c_1 + c_2\end{pmatrix}$
   We can easily solve c_1 = 1 and c_2 = -2. Therefore, $\mathbf{e_1} = 1\mathbf{v_1} - 2\mathbf{v_2}$.
3. **Use linearity to find the image of $\mathbf{e_1}$:**
   $T(\mathbf{e_1}) = T(1\mathbf{v_1} - 2\mathrm{v}_2) = 1T(\mathbf{v_1}) - 2T(\mathbf{v_2})$
   Now substitute the known images: $T(\mathbf{e_1}) = 1\begin{pmatrix}3\\3\end{pmatrix} - 2\begin{pmatrix}1\\2\end{pmatrix} = \begin{pmatrix}3\\3\end{pmatrix} - \begin{pmatrix}2\\4\end{pmatrix} = \begin{pmatrix}1\\-1\end{pmatrix}$
4. **Construct the matrix:** Now we have both T(e_1) and T(e_2).
   $\mathbf{A} = [T(\mathbf{e_1}) \quad T(\mathbf{e_2})] = \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix}$
   This is the standard matrix for the transformation T.

Method 2: (Stacking)

From the given information $T\begin{pmatrix}1\\2\end{pmatrix} = \begin{pmatrix}3\\3\end{pmatrix}$ and $T\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}1\\2\end{pmatrix}$, we form two matrices

$\mathbf{B} = [\mathbf{v_1} \quad \mathbf{v_2}] = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$ and $\mathbf{C} = [T(\mathbf{v_1}) \quad T(\mathbf{v_2})] = \begin{pmatrix} 3 & 1 \\ 3 & 2 \end{pmatrix}$.

We then have **AB = C** where **A** is the standard matrix of T:

$$\mathbf{A}\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ 3 & 2 \end{pmatrix} \rightarrow \mathbf{A} = \begin{pmatrix} 3 & 1 \\ 3 & 2 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix}.$$

Scenario 3: Geometrical description

Let T: $\mathbf{R}^3 \rightarrow \mathbf{R}^3$ be a linear transformation representing the rotation about the x-axis anticlockwise for 45° in the 3D space.

We can find the standard matrix for this transformation by determining where the standard basis vectors of $\mathbf{R}^3$ land after the rotation.

The standard basis vectors in $\mathbf{R}^3$ are: $\mathbf{e_1} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \mathbf{e_2} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \mathbf{e_3} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$.

The transformation T is a rotation of 45 degrees anticlockwise about the x-axis. Let's find the image of each basis vector.

1. Find the image of $\mathbf{e_1}$

The vector $\mathbf{e_1}$ lies on the x-axis, which is the axis of rotation. Any vector on the axis of rotation does not move during the rotation. Therefore, T($\mathbf{e_1}$) is the same as $\mathbf{e_1}$.

$$T(\mathbf{e_1}) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

This will be the first column of our standard matrix.

2. Find the image of $\mathbf{e_2}$ and $\mathbf{e_3}$

The vectors $\mathbf{e_2}$ and $\mathbf{e_3}$ lie in the y-z plane. The rotation about the x-axis will only affect their y and z coordinates, while their x-coordinate remains 0. This is effectively a 2D rotation in the y-z plane.

From the perspective of looking down the positive x-axis towards the origin, an anticlockwise rotation moves points from the positive y-axis towards the positive z-axis.

The standard 2D rotation matrix for an angle θ is: $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$

In our case, θ = 45°, so cos(45°) = √2 / 2 and sin(45°) = √2 / 2.

- **Image of $\mathbf{e_2}$ = (0, 1, 0):** In the y-z plane, this corresponds to the point (1, 0). We rotate it by 45 degrees:

  $$\begin{pmatrix} \cos(45^\circ) & -\sin(45^\circ) \\ \sin(45^\circ) & \cos(45^\circ) \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \cos(45^\circ) \\ \sin(45^\circ) \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix}$$

  These are the new y and z coordinates. The x-coordinate remains 0.

  So, $T(\mathbf{e_2}) = \begin{pmatrix} 0 \\ \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix}$. This is the second column of our matrix.

- **Image of $\mathbf{e_3}$ = (0, 0, 1):** In the y-z plane, this corresponds to the point (0, 1). We rotate it by 45 degrees:

$$\begin{pmatrix} \cos(45^\circ) & -\sin(45^\circ) \\ \sin(45^\circ) & \cos(45^\circ) \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -\sin(45^\circ) \\ \cos(45^\circ) \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix}$$

These are the new y and z coordinates. The x-coordinate remains 0.

So, $T(\mathbf{e_3}) = \begin{pmatrix} 0 \\ -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix}$. This is the third column of our matrix.

3. Assemble the Standard Matrix

The standard matrix **A** is formed by using T($\mathbf{e_1}$), T($\mathbf{e_2}$), and T($\mathbf{e_3}$) as its columns.

$$\mathbf{A} = [T(\mathbf{e_1}) \quad T(\mathbf{e_2}) \quad T(\mathbf{e_3})] = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

This is the standard matrix for a 45-degree anticlockwise rotation about the x-axis.